LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	VR Based training module on various aircraft
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	VR Based training module on various aircraft.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Operate VR module for FTL, RF & Embedded labs; Observe safety protocols; Instructor supervises simulation execution.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary laboratory domains covered in this module?
(b) Why is VR training preferred over traditional methods for these labs?
(c) List two safety precautions required before starting VR simulation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to FTL Lab VR Module • System initialization and safety checks • Basic flight dynamics simulation • Instrument panel interaction • Emergency procedure simulation	L & Ds	PPT, IB	15 min
(b)	• RF Lab VR Module Overview • Antenna radiation pattern visualization • Signal propagation simulation • Interference analysis tools • Data logging and interpretation	L & Ds	PPT, PDF	20 min
(c)	• Embedded Lab VR Module Introduction • Circuit board virtual assembly • Sensor integration simulation • Debugging virtual code errors • System performance testing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between FTL and Embedded lab simulations.
(b) How does the VR module assist in visualizing RF radiation patterns?
(c) What steps are taken to ensure safety during VR lab operations?

12. Home Assignment and Follow Up	Prepare a brief report on how the VR module concepts learned today will assist in operating the antenna trainer described in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	By using antenna trainer, observe, measure output and radiation pattern on following antenna: - Half wave simple dipole antenna, Quarter wave simple dipole antenna & Half wave folded dipole with reflector antenna.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min